

Topological Data Analysis Based Models of Evolving Higher-Order Networks

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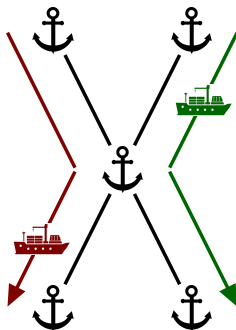


June 17, 2024

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- 1 Poisson Approximation in Weighted Random Connection Models
- 2 Topology of Higher-Order Random Connection Models
- 3 Ongoing and Future Work

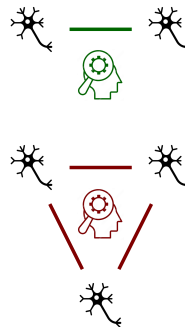
Complex Systems



(a) Maritime Traffic

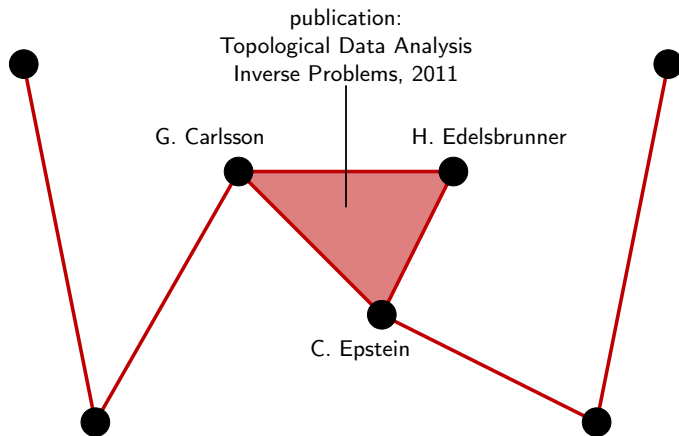


(b) Collaborations



(c) Neural Network

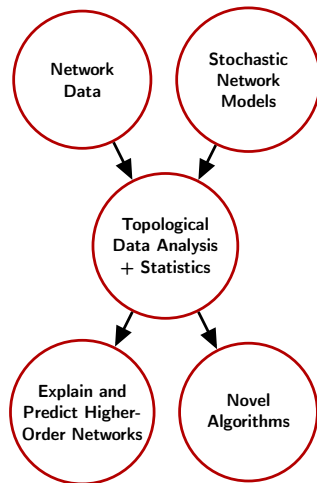
Higher-Order Networks



Simple graphs: information loss

Project Overview

- Goal: develop stochastic network models for real-world higher-order networks
- Objective 1: time-invariant features
- Objective 2: network dynamics



Spatial Random Connection Models

• Goal

- power-law distribution
- small-world property
- community structure (clustering)

• Model

- vertices: point process on a metric space
- weights: i.i.d. random variables
- edges: random connection function

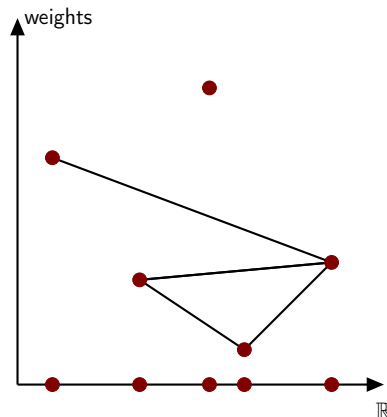


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Kernel-Based Spatial Random Network Model

- vertices: point process $\mathcal{P}_s$ on $\mathbb{R}^d$ with intensity $s > 0$
- weights: iid random variables W_x , distribution $F_W(w)$
- preferential attachment kernel $\kappa: [0, \infty) \times [0, \infty) \rightarrow [0, \infty)$

$$\kappa(w_1, w_2) := (w_1 \wedge w_2)(w_1 \vee w_2)^a \quad (a \geq 0)$$

- connection probability: $\varphi: [0, \infty) \rightarrow [0, 1]$

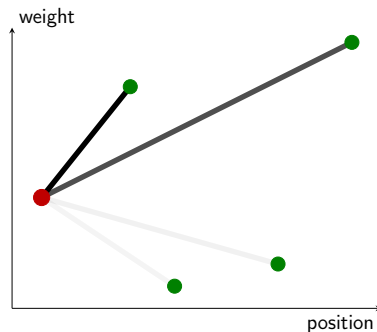
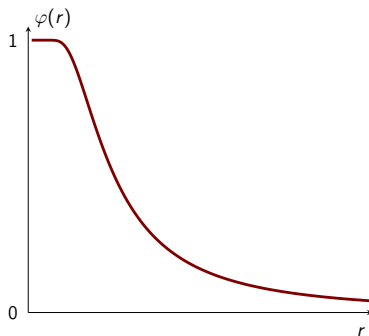
$$\varphi\left(\frac{|B_{|x-y|}(o)|}{v_s \kappa(W_x, W_y)}\right) \int_0^\infty \varphi(r) dr = 1$$

- regularity conditions on F_W and φ

$$\lim_{r \uparrow \infty} \frac{\varphi(tr)}{\varphi(r)} = t^{-\alpha} \quad \lim_{w \uparrow \infty} \frac{1 - F_W(tw)}{1 - F_W(w)} = t^{-\beta} \quad \alpha > 1, \beta \geq a\alpha$$

Example

$$\varphi(r) := 1 - \exp(-r^{-2}/\pi) \quad \kappa(w_1, w_2) := w_1 w_2 \quad v_s := 1$$



Goal of the Study

- Spatial distribution of k -degree vertices:

$$\xi_s := \sum_{x \in \mathcal{P}_s \cap [0,1]^d} \mathbb{1}\{\deg(x) = k\} \delta_x$$

- Number of degree- k vertices in $[0,1]^d$:

$$D_s := \xi_s([0,1]^d)$$

- Goal: approximate ξ_s as $s \rightarrow \infty$

Scaling Parameter

- Specify scaling v_s such that $\mathbb{E}[D_s] = 1$:

$$k! = s \mathbb{E}[(sv_s h(W))^k \exp(-sv_s h(W))]$$

$$h(w) := \mathbb{E}[\kappa(w, \bullet)] = w \mathbb{E}[\mathbb{1}\{W \geq w\} W^a] + w^a \mathbb{E}[\mathbb{1}\{W < w\} W]$$

- Note: $1/s \in o(v_s)$
- Typical degree:

$$\mathbb{E}[\deg(o)|W_o] = sv_s h(W_o) \quad \sigma_s := sv_s$$

- $1/(2s)$ -quantile of F_W :

$$\mathbb{P}(W \leq w_s) = 1/(2s)$$

Poisson Approximation

Theorem (Hirsch, Jahnel, Jhawar, J.)

If

ζ : Poisson point process with intensity $\mathbb{1}\{x \in [0, 1]^d\}$

Then

$$\lim_{s \uparrow \infty} d_{\text{KR}}(\xi_s, \zeta) = \lim_{s \uparrow \infty} \sup_{h \in \text{LIP}} \{|\mathbb{E}[h(\xi_s)] - \mathbb{E}[h(\zeta)]|\} = 0$$

Stein's Method

Bobrowski, O., et al. "Poisson process approximation under stabilization and Palm coupling." *Annales Henri Lebesgue* 5 (2022): 1489-1534.

- $(\mathbb{X}, \mathcal{X})$: lcsch space
- $N_{\mathbb{X}}$: set of σ -finite counting measures on $\mathbb{X}$
- $\mathcal{F}(\mathbb{X})$: set of closed subsets of $\mathbb{X}$
- $\mathcal{S} : \mathbb{X} \times N_{\mathbb{X}} \rightarrow \mathcal{F}$: stopping sets

$$\{\omega \in N_{\mathbb{X}} : \mathcal{S}(x, \omega) \subset A\} = \{\omega \in N_{\mathbb{X}} : \mathcal{S}(x, \omega \cap A) \subset A\} \quad (\forall A \subseteq \mathbb{X})$$
- Indicator

$$g : \mathbb{X} \times N_{\mathbb{X}} \rightarrow \{0, 1\} \quad g(x, \omega) = g(x, \omega \cap A) \quad \forall A \supset \mathcal{S}(x, \omega)$$

Stein's Method

Theorem (Bobrowski, Schulte, Yogeshwaran)

If

- η, ζ : Poisson point processes on $\mathbb{X}$ with intensities $\lambda_\eta, \lambda_\zeta < \infty$
- ξ : point process on $\mathbb{X}$ with intensity λ_ξ

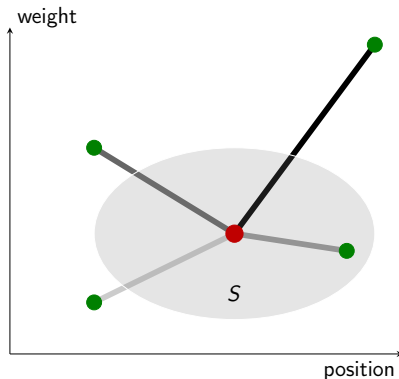
$$\xi[\eta] := \sum_{x \in \eta} g(x, \eta) \delta_x \quad \lambda_\xi = \int \mathbb{E}[g(x, \eta + \delta_x)] d\lambda_\eta < \infty$$

Then

$$d_{\text{KR}}(\xi, \zeta) \leq d_{\text{TV}}(\lambda_\xi, \lambda_\zeta) + E_1 + E_2 + E_3 + E_4$$

Proof Sketch

- Problem: g should be localized to $S \in \mathcal{F}$
- Step 1: mark approximation
neglect high weighted points
- Step 2: reach approximation:
neglect far points



Proof Sketch

Mark Approximation

- Neglect vertices with “very large marks”
- Neglect vertices with marks above w_s^η
- Two types of errors
 - all points: $\deg(x) = k$; after truncation: $\deg(x) < k$
 - all points: $\deg(x) > k$; after truncation: $\deg(x) = k$
- Superposition: split expectation into two parts

Reach Approximation

- Mecke theorem
Markov inequality
- Dominated convergence theorem

Assumptions

$$\delta := \frac{\alpha - 1}{2} \quad \exists K > 0 \quad \exists \eta \in (0, 1)$$

$$\text{A.1} \quad \sigma_s^{-1} w_s^{-\eta} \in o(1/\log(s))$$

$$\text{A.2} \quad F(w_s^\eta) w_s^{-(1+K)(1-\eta)} \in o(1/\log(s))$$

$$\text{A.3} \quad w_s^{-(1-K\delta)(1-\eta)} \in o(1/\log(s))$$

Example: Stretched Exponential Distribution

$$F(w) = p \exp(-w^{-\rho}) \quad (w \in [0, b])$$

Lemma (σ_s -scaling)

$$\sigma_s \asymp \log(s)^{1+1/\rho}$$

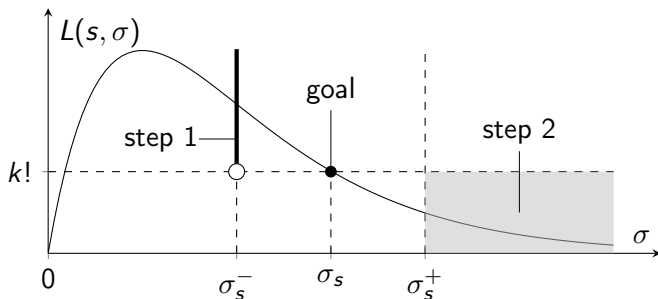
Idea: show bounds for σ_s for some $M > 0$

$$\sigma_s^- := M^{-1} \log(s)^{1+1/\rho} \quad \sigma_s^+ := M \log(s)^{1+1/\rho}$$

Scaling of σ_s

$$\sigma_s^- := M^{-1} \log(s)^{1+1/\rho} \quad \sigma_s^+ := M \log(s)^{1+1/\rho} \quad (M > 0)$$

$$L(s, \sigma) := s \mathbb{E}[(\sigma h(W))^k \exp(-\sigma h(W))]$$



Step 1

Assumption: $\sigma = \sigma_s^-$

$$\begin{aligned} L(s, \sigma_s^-) &= s \mathbb{E}[(\sigma_s^- h(W))^k e^{-\sigma_s^- h(W)}] \\ &\geq s \mathbb{E}\left[e^{-\sigma_s^- h(W)} \mathbb{1}\{\sigma_s^- h(W) \geq 1, W \leq Kw_s\}\right] \end{aligned}$$

Note that

$$\lim_{s \uparrow \infty} w_s = 0; \quad \lim_{w \downarrow 0} h(w)/w = \mu_a$$

$$\begin{aligned} \lim_{s \uparrow \infty} L(s, \sigma_s^-) &\geq \lim_{s \uparrow \infty} s \mathbb{E}\left[e^{-2\sigma_s^- W \mu_a} \mathbb{1}\left\{\frac{2}{\sigma_s^- \mu_a} \leq W \leq Kw_s\right\}\right] \\ &\geq \lim_{s \uparrow \infty} s e^{-2\sigma_s^- Kw_s \mu_a} \mathbb{P}\left(\frac{2}{\sigma_s^- \mu_a} \leq W \leq Kw_s\right). \end{aligned}$$

Step 1

- Approximate the exponential term:

$$e^{-2\sigma_s^- K w_s \mu_a} = \exp\left(-2K\mu_a \log(s)^{1+1/\rho} \log(2ps)^{-1/\rho}/M\right) \\ \asymp s^{-2K\mu_a/M}$$

- Lower bound the probability term:

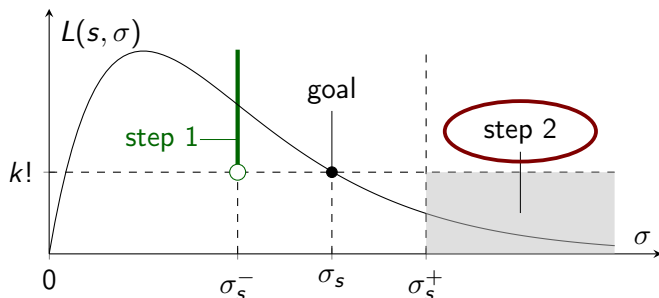
$$\mathbb{P}\left(\frac{2}{\sigma_s^- \mu_a} \leq W \leq K w_s\right) \geq \mathbb{P}(W \leq K w_s) - \mathbb{P}(W \leq K w_s/2) \\ \asymp p(2ps)^{-K^{-\rho}} - p(2ps)^{-(K/2)^{-\rho}} \asymp s^{-K^{-\rho}}$$

- Choose K and M large enough:

$$\lim_{s \uparrow \infty} L(s, \sigma_s^-) \geq \lim_{s \uparrow \infty} s^{1-K^{-\rho}-2M^{-1}K\mu_a} = \infty$$

Step 2

$$L(s, \sigma) := s \mathbb{E}[(\sigma h(W))^k \exp(-\sigma h(W))]$$



Step 2

Assumption: $\sigma \geq \sigma_s^+$

$$\begin{aligned}
 \limsup_{s \uparrow \infty} \sup_{\sigma \geq \sigma_s^+} L(s, \sigma) &= \limsup_{s \uparrow \infty} \sup_{\sigma \geq \sigma_s^+} s \mathbb{E}[(\sigma h(W))^k e^{-\sigma h(W)}] \\
 &\leq \limsup_{s \uparrow \infty} \sup_{\sigma \geq \sigma_s^+} s \mathbb{E}[C e^{-\sigma/2 h(W)}] \\
 &\leq C \limsup_{s \uparrow \infty} s \mathbb{E}[e^{-\sigma_s^+/2 h(W)}]
 \end{aligned}$$

Step 2

- $K > 0$

$$\limsup_{s \uparrow \infty} \sup_{\sigma \geq \sigma_s^+} L(s, \sigma)$$

$$\leq \liminf_{s \uparrow \infty} \left(s \mathbb{E} \left[e^{-\sigma_s^+ / 2 h(W)} \mathbb{1}_{\{Kw_s \leq W \leq b\}} \right] \right. \\ \left. + s \mathbb{P}(W \notin [Kw_s, b]) \right)$$

- $K \leq 2^{-1/\rho}$

$$w \in [Kw_s, b] \implies h(w) \geq w\mu_+(b)$$

$$\mathbb{E} \left[e^{-\sigma_s^+ / 2 h(W)} \mathbb{1}_{\{Kw_s \leq W \leq b\}} \right] \leq e^{-\sigma_s^+ / 2 Kw_s \mu_+(b)} \in o(s^{-2})$$

Assumption 1

$$\begin{aligned}\liminf_{s \uparrow \infty} \frac{\log(s)}{\sigma_s w_s^\eta} &= \liminf_{s \uparrow \infty} \frac{\log(s)}{\sigma_s \log(2ps)^{-\eta/\rho}} \\ &\leq \lim_{s \uparrow \infty} M \log(s)^{-1-1/\rho} \log(s)^{1+\eta/\rho} = 0\end{aligned}$$

Assumption 2

$$F(w_s^\eta) = p \exp(-w_s^{-\rho\eta}) = p \exp(-\log(2ps)^{1/2})$$

$$\begin{aligned} \limsup_{s \uparrow \infty} \log(s) w_s^{-(K+1)(1-\eta)} F(w_s^\eta) \\ &= p \lim_{s \uparrow \infty} \log(s) \log(2ps)^{(K+1)/(2\rho)} \exp(-\log(2ps)^{1/2}) \\ &= p \lim_{s \uparrow \infty} \log(s)^{1+(K+1)/(2\rho)} \exp(-\log(s)^{1/2}) \\ &= p \lim_{s \uparrow \infty} \exp(-\log(s)^{1/2}) = 0 \end{aligned}$$

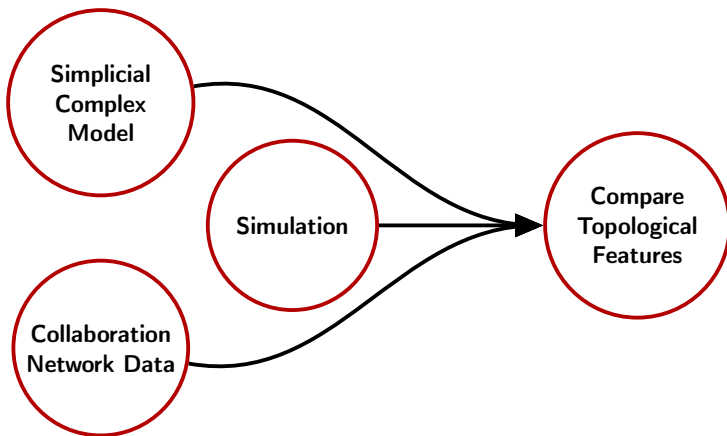
Assumption 3

$$\begin{aligned} \limsup_{s \uparrow \infty} \log(s) w_s^{(K\delta-1)(1-\eta)} \\ = \lim_{s \uparrow \infty} \log(s) \log(2ps)^{-(K\delta-1)/(2\rho)} = 0. \end{aligned}$$

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Outline



Age-Dependent Random Connection Model

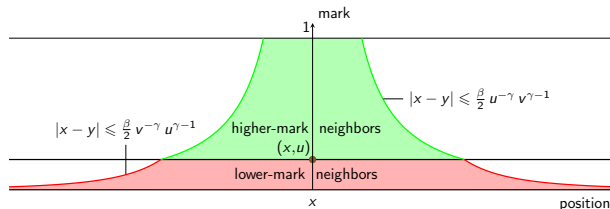
Gracar, P., et al. "The age-dependent random connection model." *Queueing Systems* 93 (2019): 309-331.

Vertices: Poisson process

Connections: $\{(x, u), (y, v)\} \in \mathcal{P}^2$

$$\mathcal{P} \subset \mathbb{R} \times [0, 1]$$

$$|x - y| \leq \frac{\beta}{2} u^{-\gamma} v^{\gamma-1} (u \leq v)$$

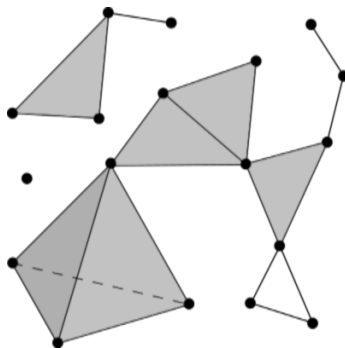


Simplicial Complex

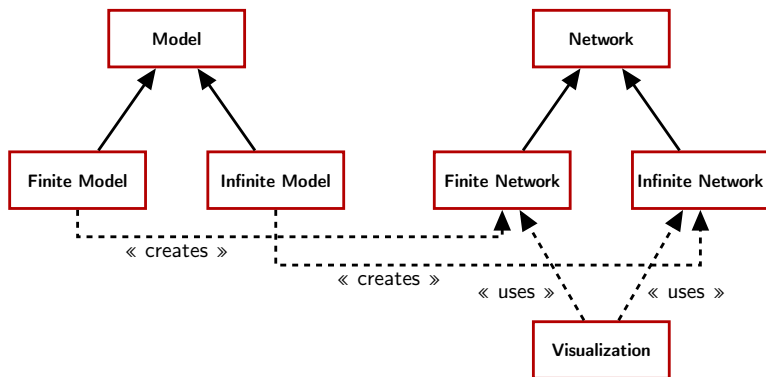
Idea: Consider ADRCM as a “clique complex”

Simplicial Complex

- (P, Σ)
- $P \neq \emptyset, |P| < \infty$
- $\Sigma \subseteq \mathcal{P}(P)$
- $p \in P \implies \{p\} \in \Sigma$
- $\sigma \in \Sigma, \Delta \subset \sigma \implies \Delta \in \Sigma$



Simulation

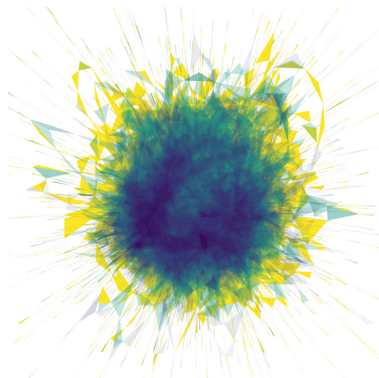


- Finite-Size Effects
- Conjectures

- **C++:** 4 149 SLoC
- **Python:** 6 654 SLoC

Network Data

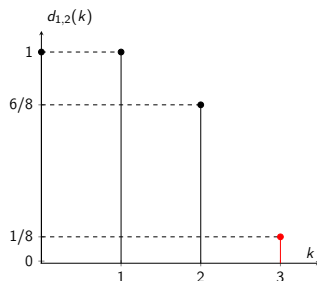
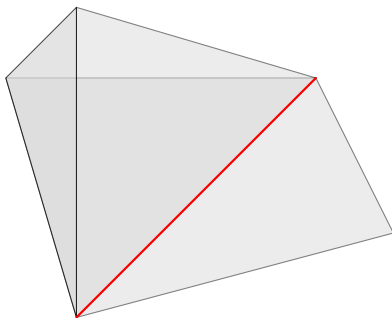
- Collaboration network data
- Source: arXiv preprint server
- Vertices: $\sim 8 \cdot 10^5$ authors
- Simplices: $\sim 10^6$ documents



Higher-Order Degree Distributions

$$\deg_{m'}(\Delta) := |\{\sigma \in \Sigma : \sigma \supseteq \Delta, |\sigma| = m' + 1\}|$$

$$d_{m,m'}(k) := \mathbb{P}(\deg_{m'}(\Delta) \geq k)$$



Higher-Order Degree Distributions

Theorem (Hirsch, J.)

If

$$\deg_{m'}(\Delta) := |\{\sigma \in \Sigma : \sigma \supseteq \Delta, |\sigma| = m' + 1\}|$$

$$d_{m,m'}(k) := \mathbb{P}(\deg_{m'}(\Delta_m) \geq k)$$

Then

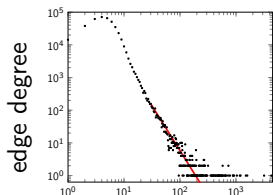
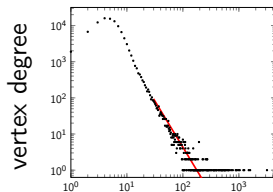
$$\lim_{k \uparrow \infty} \log(d_{m,m'}(k)) / \log(k) = m - (m + 1)/\gamma$$

Proof Steps.

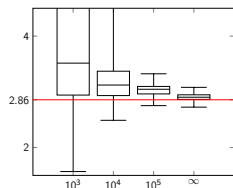
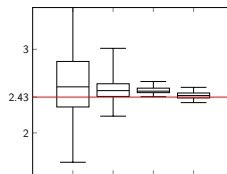
- Define Palm distribution
- Lower bound: find $B_k \subset \mathbb{R} \times [0, 1]$ in which $m + 1$ points surely connect
- Upper bound: reduce to $m' = m + 1$



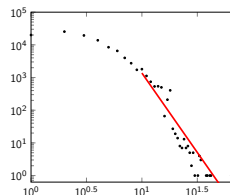
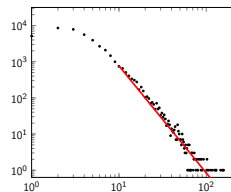
Parameter Fitting



(a) Simulation

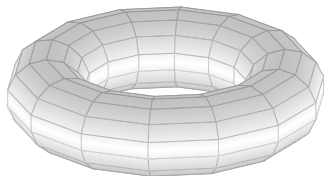


(b) Convergence



(c) Data and Model

CLT for Betti Numbers



$$\beta_0 = 1; \beta_1 = 2; \beta_2 = 1$$

Theorem (Hirsch, J.)

If $W_n = [-n/2, n/2] \times [0, 1]$,
 $0 < \gamma < 1/4$, $q \geq 0$, then

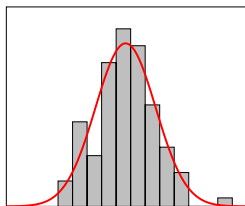
$$\frac{\beta_q^{(n)} - \mathbb{E}[\beta_q^{(n)}]}{\sqrt{\text{Var}(\beta_q^{(n)})}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Proof Sketch.

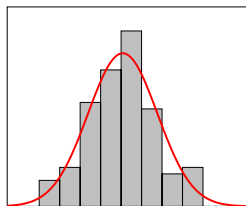
- Add one cost operator: $\delta(\mathcal{P}, (0, u)) := \beta(\mathcal{P} \cup \{(0, u)\}) - \beta(\mathcal{P})$
- Weak stabilization: $\lim_{n \uparrow \infty} \delta(\mathcal{P} \cap W_n, (0, u)) < \infty$
- Moment bound: $\sup_{n \geq 1} \mathbb{E}[\delta(\mathcal{P} \cap W_n, (0, u))^4] < \infty$



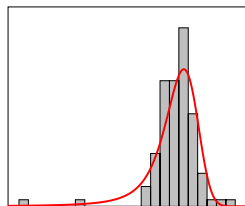
Simulated Distribution of Betti Numbers



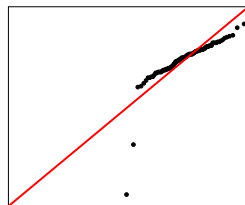
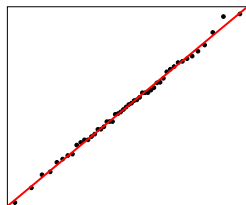
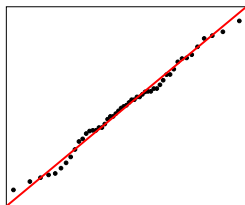
(a) $\gamma = 0.25$



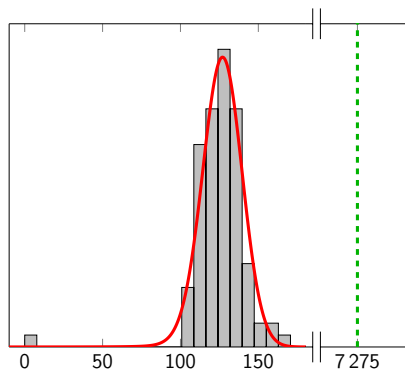
(b) $\gamma = 0.50$



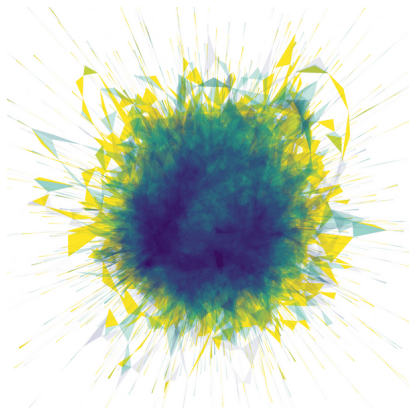
(c) $\gamma = 0.67$



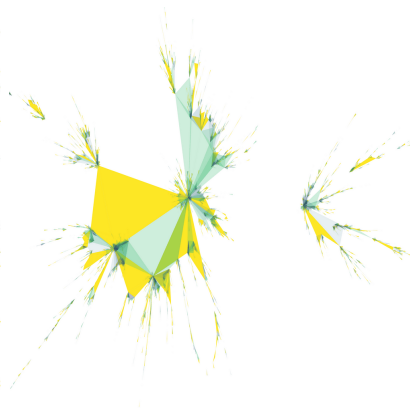
Hypothesis Test



Hypothesis Test



(a) Network Data



(b) Simulation

Asymptotics of Edge Counts

- $S_n := |\{(x, u), (y, v)) \in \mathcal{P}^2 : (x, u) \leftrightarrow (y, v), x \in [0, n]\}|$

Theorem (Hirsch, J.)

Central Limit Theorem

- $0 < \gamma < 1/2$
- $n^{-\frac{1}{2}}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Stable Limit Law

- $1/2 < \gamma < 1$
- $n^{-\gamma}(S_n - \mathbb{E}[S_n]) \xrightarrow{d} \mathcal{S}(1/\gamma)$

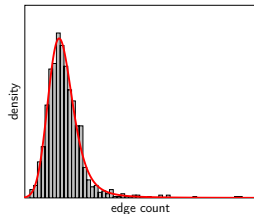
Proof Sketch.

- Harris-FKG theorem
- Upper bound sum of covariance terms
- Apply CLT (associated rvs)
- Neglect high-mark vertices
- Approximate low-mark vertices as iid Pareto rvs

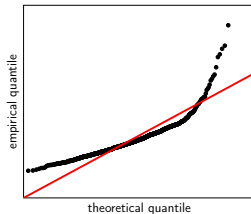


Stable Distribution of Edge Counts

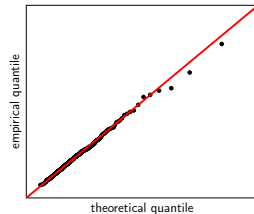
$\gamma = 0.6$: Stable Distribution



(a) Edge Counts



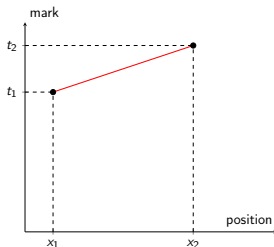
(b) Normal Distribution



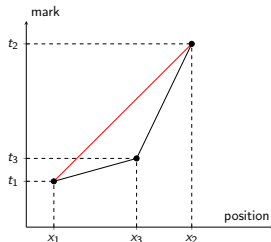
(c) Stable Distribution

Thinned Model

- **Goal:** match the $d_{0,1}$ and $d_{1,2}$ exponents separately (if $n \rightarrow \infty$)
- **Observation:** high edge degree $\implies$ both endpoints are old
- **Solution:** remove edges that do not affect exponent of $d_{1,2}$



Protected: $t_1 \approx t_2$



Protected: $t_1 \approx t_3$

- Remove exposed edges independently with probability $1 - t_1^\eta$
- $\lim_{k \uparrow \infty} d_{0,1}(k) \sim k^{-\frac{1}{\eta-\gamma}}$ $\lim_{k \uparrow \infty} d_{1,2}(k) \sim k^{1-\frac{2}{\gamma}}$

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- 2 Topology of Higher-Order Random Connection Models
- 3 Ongoing and Future Work

Bipartite Random Connection Model

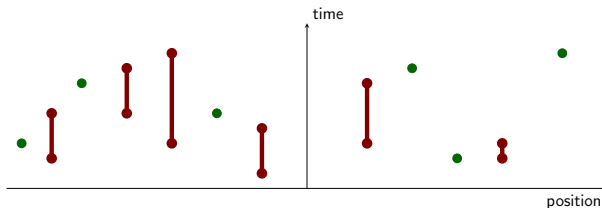
- Higher-order ADRCM: lack of interaction strength
- Idea: build a bipartite graph: vertices and interactions
- Two Poisson point processes $\mathcal{P}, \mathcal{P}' \subset \mathbb{R} \times [0, 1]$
- Connections: $\mathcal{P} \ni (x, u) \leftrightarrow (y, v) \in \mathcal{P}'$ if

$$|x - y| \leq \beta u^{-\gamma} v^{-\gamma'}$$

- Simplices: simplices in the Dowker complex

$$\Sigma = \{ \Delta \subseteq \mathcal{P} : \exists p' \in \mathcal{P}' : \forall p \in \Delta, p \leftrightarrow p' \}$$

Structural Dynamics of RCMs



- Birth-Death processes for vertices
- Show functional limit theorems
- Explain dynamic aspects of higher-order networks

Summary

- **Complex Systems**

- Crucial multibody relationships
- Higher-order networks, simplicial complexes

- **Goal**

- Stochastic models for higher-order networks
- Time-invariant and time-varying structures

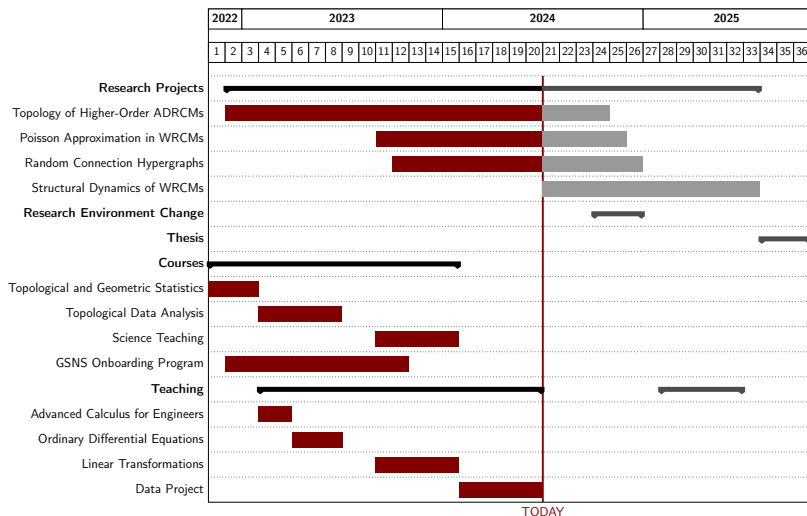
- **Research Projects**

- Poisson Approximation in WRCMs
- Topology of Higher-Order ADRCMs

- **Future Work**

- Random Connection Hypergraphs
- Structural Dynamics of RCMs

PhD Plan



Thank you for your attention!

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